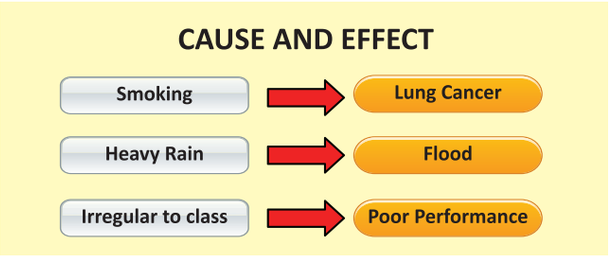
There is a high positive correlation between test-1 and test-2. That is those who perform well in test-1 will also perform well in test-2 and those who perform poor in test-1 will perform poor in test- 2.

The students can also verify the results by using shortcut method.

4.4.3 Limitations of Correlation

Although correlation is a powerful tool, there are some limitations in using it:

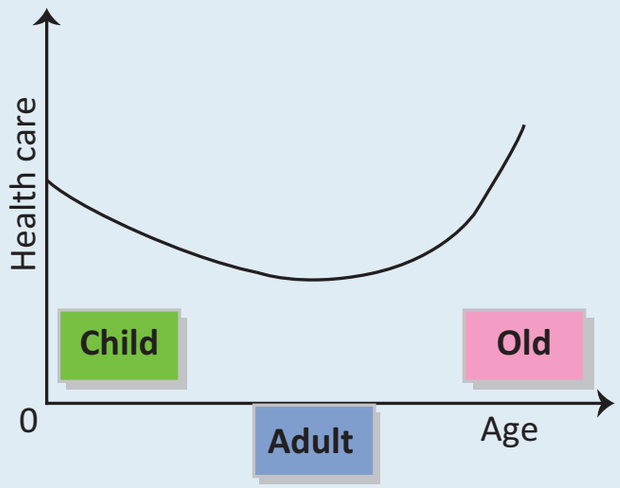


1. Outliers (extreme observations) strongly influence the correlation coefficient. If we see outliers in our data, we should be careful about the conclusions we draw from the value of r. The outliers may be dropped before the calculation for meaningful conclusion.
2. Correlation does not imply causal relationship. That a change in one variable causes a change in another.

NOTE

1. **Uncorrelated** : Uncorrelated ($r = 0$) implies no 'linear relationship'. But there may exist non-linear relationship (curvilinear relationship).

   Example: Age and health care are related. Children and elderly people need much more health care than middle aged persons as seen from the following graph.



However, if we compute the linear correlation r for such data, it may be zero implying age and health care are uncorrelated, but non-linear correlation is present.

2. **Spurious Correlation** : The word '**spurious**' from Latin means **'false'** or 'illegitimate'. *Spurious correlation means an association extracted from correlation coefficient that may not exist in reality.*

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